

ACIT5910 Master's Thesis, Phase 1

Beyond the Weakest Link: The Skepticism Paradox Why Phishing Works Despite Rising Skepticism

A Mixed-Methods Study of Loneliness, Social Engineering,
and Digital Trust in Norway

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ACIT5910 Phase 1 criteria mapped: Individual text 5,000–10,000 words · Introduction + Literature Review · Problem statement and research questions · Plan for Phase 2–3 research · APA references · Times New Roman 12 pt · 1.5 line spacing

Abstract. This Phase 1 submission investigates why phishing remains effective despite rising public scepticism toward unknown senders. The thesis introduces the *scepticism paradox*: general caution and lure-specific vulnerability can coexist when phishing imitates social belonging, institutional authority, or emotionally meaningful opportunity rather than obvious stranger fraud. Moving beyond the weakest-link narrative, the study plans mixed-methods research in Norway examining loneliness, situational pressure, verification habits, and security culture. The submission comprises Introduction and Literature Review (ACIT5910), with problem statement, research questions, and a plan for empirical work in Phase 2 and Phase 3.

Keywords: phishing, scepticism paradox, loneliness, social engineering, human factors, Norway

Introduction

Background and motivation

Phishing remains one of the most common entry points to serious cyber incidents, despite decades of technical investment and rising public awareness. Organisations deploy email filters, multi-factor authentication, security awareness programmes, and incident response teams. Yet social engineering attacks continue to succeed at scale (European Union Agency for Cybersecurity [ENISA], 2024; Cybersecurity and Infrastructure Security Agency [CISA], n.d.). This persistence suggests that the dominant explanation for phishing success—that users are the "weakest link" in an otherwise secure system—is incomplete.

In Norway, citizens are among the most digitalised populations in Europe. BankID, ID-porten, Altinn, and digital mailboxes structure everyday life. Public campaigns from Finans Norge, the Norwegian National Security Authority (NSM), and the Norwegian Communications Authority (Nkom) repeatedly advise scepticism toward unknown senders and unexpected links (Finans Norge, 2026; NSM, 2026). Awareness is rising. Reported phishing attempts remain high. This thesis begins from that gap: phishing works despite rising scepticism, not because scepticism is absent.

The proposed explanation is the scepticism paradox. General caution toward strangers may coexist with heightened openness to messages that appear socially or institutionally legitimate. Phishing often imitates belonging, safety, authority, or opportunity rather than obvious foreign fraud. The thesis therefore investigates emotional, cognitive, and contextual factors that make individuals vulnerable to this form of social engineering in Norway, with primary data collection planned among adults in higher education.

The relevance of this problem extends beyond individual victims. Organisational breaches that begin with a single click can expose thousands of records, disrupt public services, and undermine trust in digital infrastructure. For students and staff in higher education, phishing may target credentials, research data, fee payments, and personal information simultaneously. A human-centred explanation of

why cautious people still respond to credible lures is therefore both academically timely and practically necessary for institutions that cannot rely on technical controls alone.

Beyond the weakest link

Cybersecurity discourse frequently treats human error as the terminal explanation for phishing incidents. When an employee clicks a malicious link, training materials and post-incident reviews often conclude that the user failed to follow policy. When a technical control fails, organisations analyse system design. This asymmetry individualises a structural problem (Hadnagy, 2018).

The weakest-link metaphor assumes that more scepticism uniformly reduces risk. If that were true, phishing prevalence should fall as public awareness rises. Norwegian statistics do not support a simple linear relationship: NSM received more than 100,000 reports of phishing attempts in 2024, and the 2026 national risk assessment emphasises increasing sophistication in social manipulation, including AI-assisted content (NSM, 2026; Nkom, n.d.).

Attackers do not primarily exploit ignorance. They exploit context: urgency, authority, social relevance, and emotional need (Parker & Flowerday, 2020; Tsai et al., 2021). A person may reject a crude lottery scam yet respond to a message impersonating a university IT desk during exam season. This thesis treats phishing as a human decision problem disguised as an email, not as a failure of intelligence or attention alone.

The weakest-link frame also carries ethical weight. When organisations describe users as the predictable failure point, they may underinvest in safer system defaults, clearer verification workflows, and non-punitive reporting channels. Blame-oriented narratives can discourage disclosure of near misses, reducing the data available for organisational learning (Chen et al., 2024). Moving beyond the weakest link is therefore not a rhetorical preference but an analytical requirement for understanding phishing as a socio-technical phenomenon.

The scepticism paradox

The scepticism paradox names a specific tension at the centre of this project. Individuals may report high general scepticism toward

unknown or unsolicited digital contact while remaining vulnerable to phishing that mimics peers, institutions, employers, or emotionally meaningful opportunities.

Public guidance emphasises distrust of strangers (Finans Norge, 2026; Nettvett.no & NSM, n.d.). Contemporary phishing increasingly impersonates trusted domains of life: banking, taxation, streaming services, study administration, and social connection. The lure is lure-specific: scepticism that protects against stranger-framed scams may not protect against institutionally or socially framed ones (Klüttsch et al., 2024; Parker & Flowerday, 2020).

Loneliness adds a second axis. Loneliness is not merely the absence of company but a signal of unmet social connection needs that can increase openness to contact and reassurance (Cacioppo & Patrick, 2008). Among younger adults, fear of missing out (FOMO), platform exposure, and impulsivity may interact with phishing cues differently than among older cohorts studied in classic fraud research (Przybylski et al., 2013; Klüttsch et al., 2024). The paradox therefore links rising social disconnection and rising general caution—two trends visible in Norwegian public discourse (Statistics Norway [SSB], 2025; Khrono & Norwegian Student Organisation, 2024).

Formally stated, the scepticism paradox holds that protective caution toward strangers and vulnerability to socially or institutionally credible lures can coexist in the same individual. This contradicts training models that treat scepticism as a single protective trait. It also explains why digitally literate populations remain exposed: literacy helps recognise crude fraud but does not automatically protect against messages that fit existing needs for belonging, compliance, or opportunity.

Problem statement and research questions

Problem statement: Phishing remains effective in a context characterised by increased general online scepticism and measurable social disconnection, because credible impersonation exploits legitimate human needs rather than ignorance alone. Existing training models that treat users as uniformly weak links do not adequately explain lure-specific vulnerability.

Main research question: How can phishing remain effective despite rising scepticism, and which emotional, cognitive, and contextual

factors make individuals especially vulnerable beyond the weakest-link explanation?

Sub-questions: (1) How do loneliness, social connection needs, and general online scepticism relate to phishing susceptibility? (2) Which emotional and situational factors—urgency, authority, fear, shame, stress—most strongly influence unsafe responses? (3) What is the gap between knowing about phishing and acting safely at the moment of decision? (4) Which protective factors—training, culture, verification routines, peer support—are experienced as most effective?

These questions are designed to be answerable within a mixed-methods master thesis while remaining open enough to accommodate unexpected findings from interviews. They deliberately avoid deterministic claims about loneliness causing clicks; instead, they examine relationships, interactions, and lived experience under varying lure conditions.

Scope and delimitations

The study focuses on phishing, smishing, vishing, and impersonation scams involving emotional or social manipulation. The geographic frame is Norway, with international literature as theoretical backdrop. Primary recruitment is planned among students and staff in higher education (emphasis 18–35 years), while the conceptual argument applies more broadly to digitally active adults.

Malware reverse engineering, ransomware operations, DDoS, and deep technical forensics are out of scope as main topics. The thesis does not claim that loneliness causes phishing in a deterministic sense; loneliness is treated as one measurable variable among others in a mixed-methods design planned for Phase 2 and Phase 3.

The study will not deploy live deceptive phishing against participants without explicit ethical approval. Scenario vignettes present hypothetical messages and measure intended behaviour rather than actual credential compromise. This design balances ecological validity with participant welfare and institutional ethics requirements.

Contribution

Theoretically, the thesis offers a lure-specific model of phishing vulnerability centred on the scepticism paradox, extending Protection Motivation Theory (PMT) where threats are not initially recognised as

such (Rogers, 1975; Floyd et al., 2000). Empirically, it plans mixed-methods evidence from Norway using scenario vignettes, validated loneliness items, and semi-structured interviews (Braun & Clarke, 2006). Practically, it aims to inform lure-specific awareness design, non-punitive reporting culture, and integration of student welfare with cybersecurity communication (Chen et al., 2024).

Norwegian studies by Tjostheim and Waterworth (2020, 2022) demonstrate that cognitive reflection predicts phishing susceptibility in representative samples. This thesis asks whether emotional and social variables add explanatory power beyond cognitive style—particularly for socially and institutionally framed lures that dominate contemporary campaigns (ENISA, 2024; Trend Micro, 2026).

The thesis also contributes a named analytical concept—the scepticism paradox—that can be tested, refined, or rejected through empirical work. Whether or not all hypotheses are supported, the paradox provides a structured way to discuss why rising awareness has not eliminated phishing harm in high-trust digital societies.

Structure of the Phase 1 submission and planned thesis

This Phase 1 submission (ACIT5910) comprises the Introduction and Literature Review, following course requirements of 5,000–10,000 words with a clear problem statement, research questions, and a plan for research in Phase 2 (ACIT5920) and Phase 3 (ACIT5930). Subsequent phases will add methodology, results, discussion, and conclusions. Figure 1 (planned) will summarise the scepticism paradox as a conceptual model linking rising stranger-scepticism, loneliness, lure type, and protective factors.

ACIT5910 also requires draft submission, peer review participation, and a process memo documenting revisions. The writing workshop learning outcomes emphasise clarity, coherence, audience awareness, and academic integrity. This document is written for an academic supervisor and examiner audience in cybersecurity, with APA 7 references and no unsupported personal opinion in the literature review sections.

Planned research for Phase 2 and Phase 3

Phase 2 will implement an online survey (target N = 120–180) combining loneliness measures, general scepticism items, situational

pressure scales, and six scenario vignettes (S1–S6) manipulating lure framing (stranger, social, institutional, authority, opportunity, belonging). Phase 3 will conduct 12–18 semi-structured interviews and integrate findings through reflexive thematic analysis. Ethics approval via the Norwegian Agency for Shared Services in Education and Research (SIKT) is planned before data collection. A pilot with 10–15 OsloMet participants will test completion time and item clarity.

Four hypotheses guide the empirical work: H1—higher loneliness associates with higher susceptibility for socially framed lures; H2—general scepticism reduces stranger-framed but not institutional/social lures (interaction); H3—situational pressure partially mediates loneliness and unsafe behaviour; H4—verification habits and non-punitive security culture reduce susceptibility (PMT coping appraisal).

Integration in Phase 3 will use joint displays connecting statistical patterns with interview quotes on credibility, shame, and reporting culture. If quantitative and qualitative findings diverge—for example, if loneliness correlates weakly in survey data but emerges strongly in narratives—the thesis will report that tension transparently rather than forcing a single story.

Literature Review

Phishing and social engineering in cybersecurity research

Phishing is a social engineering attack in which victims are manipulated into unsafe actions: clicking links, revealing credentials, approving transactions, or downloading malware. Success depends on perceived credibility, relevance, and emotional meaning rather than technical sophistication alone (Hadnagy, 2018; Conteh & Schmick, 2016). ENISA (2024) and Microsoft (2024) consistently rank social engineering among the dominant initial access vectors in organisational incidents.

Human factors research shows that susceptibility correlates with persuasion strategies, workload, time pressure, and perceived sender legitimacy more strongly than with intelligence (Parker & Flowerday, 2020; Tsai et al., 2021; Bayl-Smith et al., 2024). Employees and students may articulate correct security rules yet still click when messages appear urgent or socially relevant. This knowledge-behaviour gap is central to the scepticism paradox.

Frauenstein and Flowerday (2020) model personality and information-processing pathways on social network sites, suggesting that rapid social cue processing can override fraud indicators. Parker and Flowerday (2020) extend this to organisational contexts where role expectations and email volume increase verification costs. Together, these studies support contextual rather than deficit models of vulnerability.

Attack chains frequently begin with a human action even when the ultimate goal is ransomware or data exfiltration. Analysing phishing therefore remains important even when technical teams focus on later-stage malware. Initial compromise often depends on a message that does not appear malicious at the moment of decision—a condition the scepticism paradox is designed to explain.

Critique of the weakest-link narrative

The weakest-link metaphor has shaped decades of security awareness training. It implies that organisational security equals the minimum

level of user compliance. Critics argue that this framing obscures attacker strategy and organisational responsibility (Hadnagy, 2018). Attackers craft role-specific lures; systems generate high volumes of legitimate and fraudulent contact; remote work reduces informal verification opportunities (Jensen et al., 2024).

When shame and blame follow reported mistakes, near misses go unreported and organisational learning stalls (Chen et al., 2024). Bayl-Smith et al. (2024) show that protection motivation in organisational contexts depends on coping appraisal—including whether reporting is safe—rather than threat fear alone. Treating scepticism as a single trait therefore mis-specifies the problem: the relevant question is scepticism toward which lure types under which conditions.

Simulated phishing programmes can increase awareness but may also reinforce shame if failures are punished or publicised without support. Chen et al. (2024) find that training including group discussion and role-playing improves support-seeking and reporting intentions more than fear-based simulations alone. This suggests that culture and coping appraisal belong in the same analytical frame as individual traits.

From a design perspective, the weakest-link narrative encourages additive training—more modules, more reminders—without changing the decision environment. High email volume, fragmented attention, and ambiguous sender identities are structural features that increase verification cost. The scepticism paradox reframes the problem: users are not uniformly weak; they are selectively vulnerable when lures align with legitimate needs and when verification is costly.

Loneliness, belonging, and fraud susceptibility

Loneliness research describes social disconnection as a distress signal indicating unmet affiliation needs (Cacioppo & Patrick, 2008). The World Health Organization (2023) and national health authorities now treat social connection as a public-health priority. Fraud research among older adults long associates loneliness with telemarketing and scam vulnerability (Alves & Wilson, 2008; Wen et al., 2022).

Experimental work suggests loneliness can increase susceptibility to misleading offers in middle-aged and older adults (Wen et al., 2024).

Among younger adults, mechanisms may differ. Klütsch et al. (2024) find that known senders and FOMO affect phishing susceptibility on

social media. Przybylski et al. (2013) define FOMO as a pervasive concern that others might have rewarding experiences from which one is absent—a state that may amplify opportunity-framed lures. Popovac et al. (2020) show that online risk perception interacts with platform familiarity and emotional state.

In Norway, student loneliness is documented at scale. SHoT 2022 reports that 29% of students often miss someone to be with; campus presence correlates with lower loneliness (SHoT Study, 2022). SSB (2025) finds elevated loneliness among young adults in quality-of-life surveys. Yet loneliness is rarely integrated into cybersecurity curricula or awareness campaigns, creating a research and practice gap this thesis addresses.

Importantly, loneliness in this thesis is operationalised as subjective social disconnection rather than objective isolation. A person may have frequent digital contact yet feel socially disconnected—a pattern common among students during exam periods or remote study. This distinction matters because phishing lures promising belonging or community may resonate with subjective need even when objective contact is high.

The scepticism paradox in empirical perspective

Empirical support for lure-specific scepticism comes from multiple domains. Klütsch et al. (2024) demonstrate differential responses to peer- versus stranger-framed phishing on social platforms. Parker and Flowerday (2020) show that legitimacy cues and workload predict compliance even among informed users. Norwegian cognitive-reflection studies (Tjostheim & Waterworth, 2020, 2022) predict resistance to generic phishing but do not test emotionally framed institutional impersonation.

The paradox predicts an interaction: general scepticism negatively correlates with stranger-framed susceptibility but may not predict responses to BankID, Altinn, or university IT impersonation—precisely the lures emphasised in Norwegian fraud warnings (Finans Norge, 2026; Politiet, 2024). Knowing that fraud exists does not guarantee safe action when a message fits an existing need for belonging, safety, or administrative compliance (Tsai et al., 2021).

Training materials that show only crude stranger scams may inadvertently teach a mental model in which "real" phishing looks

foreign or absurd. When institutional impersonation arrives with correct branding, domestic language, and plausible context, sceptical individuals may still hesitate to dismiss the message—especially under time pressure. The paradox therefore has direct implications for awareness design.

Protection Motivation Theory and its limits

PMT explains protective behaviour through threat appraisal and coping appraisal (Rogers, 1975; Floyd et al., 2000). Meta-analytic evidence supports PMT across health and security domains (Floyd et al., 2000). In cybersecurity, PMT has been applied to policy compliance and phishing warnings (Bayl-Smith et al., 2024).

PMT is useful but incomplete for phishing. It assumes threat recognition; phishing often succeeds when messages appear benign or beneficial. Emotional needs may reframe a message as opportunity rather than risk. This thesis therefore uses PMT primarily to analyse verification habits, reporting behaviour, and perceived organisational support—coping appraisal—rather than to assume fear appeals alone will prevent clicks (Chen et al., 2024).

Threat appraisal may fail when severity and vulnerability are not perceived. A fake internship offer may not trigger fear because it appears beneficial. Coping appraisal—whether the individual believes verification is feasible and reporting is safe—may better predict behaviour for institutional lures. H4 in the planned empirical work tests this extension of PMT.

Norwegian digital trust and threat landscape

Norway combines high institutional trust with high digital dependence. NSM (2026) reports that phishing and social manipulation remain leading citizen-facing threats. Nkom documents persistent digital fraud patterns (Nkom, n.d.). Halcyon and industry reports note AI-assisted personalisation lowering the cost of credible lures (Halcyon, 2026; Trend Micro, 2026), though emotional leverage remains decisive: AI improves production; human needs determine uptake.

Public literacy campaigns improve stranger-scepticism but may inadvertently reinforce a mental model in which "real" phishing looks foreign or absurd. Contemporary campaigns impersonate domestic

institutions citizens rely on daily. The scepticism paradox is therefore particularly salient in high-trust digital societies such as Norway.

BankID and national digital identity infrastructure create strong expectations of official-looking communication. Attackers exploit those expectations by mimicking login flows, payment requests, and administrative notices. Citizens trained to distrust unknown senders may still trust messages that appear to originate from systems they use weekly. This institutional trust is a societal asset with a security shadow side.

Smishing, vishing, and channel expansion

Phishing is not limited to email. Smishing (SMS phishing) and vishing (voice phishing) use the same psychological levers with different channel constraints. SMS messages are short, urgent, and often read on mobile devices where verification is harder. Voice calls can exploit authority and real-time pressure, reducing time for reflection.

In Norway, fraud warnings increasingly cover multiple channels simultaneously (Finans Norge, 2026; NSM, 2026). The scepticism paradox applies across channels: stranger-framed SMS scams may be easier to reject than messages appearing to come from a parcel service, bank, or university administration. Channel expansion increases the number of contexts in which lure-specific vulnerability may appear.

Higher-education populations are heavy users of mobile messaging for study logistics, social coordination, and two-factor authentication. This dual use—security and social life on the same device—may blur boundaries between trusted and suspicious contact. The planned vignettes include channel-neutral framing but allow open-text responses about preferred verification methods per channel.

Knowledge-behaviour gap and decision moments

A recurring finding in human factors research is the gap between knowing security rules and acting on them under pressure (Tsai et al., 2021; Parker & Flowerday, 2020). Students may pass awareness quizzes yet report near misses during exam weeks when administrative email volume spikes. The gap is not hypocrisy; it reflects competing demands on attention, emotion, and time.

The moment of decision is often seconds long. Verification via a separate channel—calling IT, checking an official website, asking a

peer—requires additional effort that situational pressure discourages. Loneliness may increase the appeal of messages that offer immediate social or administrative resolution without friction.

Interview data in Phase 3 will explore this gap narratively: what participants knew, what they felt, and what they did. Understanding decision moments is essential for recommendations that go beyond repeating rules users already know.

Security awareness training: limitations and alternatives

Standard awareness training often emphasises identifying suspicious indicators: poor grammar, foreign domains, unexpected attachments. These heuristics are less reliable when AI improves language quality and when attackers use compromised legitimate accounts or plausible subdomains (Halcyon, 2026; Trend Micro, 2026).

Lure-specific training addresses institutional and social impersonation explicitly: how BankID requests should appear, how university IT communicates, when to verify via separate channels. Finans Norge and Nettvett materials move in this direction, but integration with student welfare and loneliness-aware communication remains limited (Finans Norge, 2026; Nettvett.no & NSM, n.d.).

Non-punitive reporting culture is a training complement, not a substitute. If reporting a near miss leads to shame or disciplinary action, awareness knowledge cannot translate into organisational learning (Chen et al., 2024; Bayl-Smith et al., 2024). H4 tests whether perceived supportive culture correlates with safer intended behaviour across vignettes.

Theoretical positioning and related frameworks

Beyond PMT, several frameworks touch parts of the scepticism paradox without naming it. Cialdini's influence principles—authority, scarcity, social proof, and liking—explain why specific message features increase compliance (Hadnagy, 2018). The thesis does not treat these as competing theories but as complementary descriptions of lure mechanics that the paradox integrates under a single question: why does general stranger-scepticism fail to generalise across lure types?

Risk communication research distinguishes perceived risk from behavioural response—a gap familiar in health and environmental

domains. Cybersecurity has imported similar ideas through human factors and usable security, yet institutional training often still equates awareness with protection. By foregrounding lure-specific scepticism and loneliness as measurable variables, this thesis makes the gap testable rather than anecdotal.

Scandinavian trust research documents high confidence in public institutions. That trust is socially valuable but creates predictable impersonation targets: BankID, Altinn, NAV, and university administration. The scepticism paradox is therefore not only a psychological claim but a socio-institutional one about how trust structures create attack surfaces that stranger-focused campaigns do not fully address.

Research gap and synthesis

Three gaps motivate this thesis. First, loneliness is well studied in older fraud victims but less systematically in Norwegian higher-education contexts. Second, general scepticism is measured in awareness surveys but rarely as an interaction with lure type. Third, the weakest-link narrative persists in practice despite human-factors evidence for contextual vulnerability.

The literature supports a combined model: phishing susceptibility emerges from emotional relevance of lure, situational pressure, unmet social need, verification opportunity, and security culture—not from individual ignorance alone. The scepticism paradox provides the original analytical lens integrating these streams. Planned mixed-methods work will test whether loneliness and lure-specific scepticism jointly predict intended unsafe behaviour beyond cognitive style alone (Tjostheim & Waterworth, 2020; Klütsch et al., 2024).

Figure 1 (to be developed in Phase 2) will present the combined conceptual model: stranger-scepticism and loneliness as parallel inputs; lure type and situational pressure as moderators; verification habits and non-punitive culture as protective outputs. This model is deliberately human-centred. It does not deny technical controls but positions them as one layer in a system where meaning, not only malware, drives initial compromise.

Remote work, cognitive bias, and reporting culture

Remote and hybrid study and work environments reduce opportunities for informal verification. A quick question to a colleague—"Did you really send this?"—is harder when communication is asynchronous and fragmented (Jensen et al., 2024). Students may receive fraudulent messages about exams, fees, or housing during periods when administrative email volume is already high. Under cognitive load, verification routines are skipped even by knowledgeable users (Tsai et al., 2021).

Authority bias, scarcity effects, and optimism bias interact with emotional states. Authority bias leads individuals to comply with apparent officials. Scarcity and urgency encourage action before verification. Optimism bias supports beliefs that fraud happens to others. Loneliness may increase the salience of socially rewarding messages; stress may reduce cognitive resources for checking sender identity (Cacioppo & Patrick, 2008).

Reporting culture matters independently of awareness. Chen et al. (2024) show that training including support-seeking and role-playing improves reporting intentions more than fear-based simulations alone. Organisations that punish reported mistakes create incentives to hide near misses. From a PMT perspective, coping appraisal includes whether help is available after a mistake (Bayl-Smith et al., 2024). Non-punitive culture is therefore treated as a protective factor in the planned empirical model.

Artificial intelligence and the production of credible lures

Industry reports document AI-assisted phishing that improves grammar, personalisation, and translation (Halcyon, 2026; Trend Micro, 2026). Lower production cost may increase campaign volume and local language quality. However, AI does not replace emotional leverage. A perfectly written fake BankID message still fails if the recipient has no reason to trust or fear its content. AI changes supply; human needs determine demand.

For the scepticism paradox, AI may widen the gap between stranger-scams and institutional impersonation. Stranger-scams with poor language are easier to dismiss; AI-generated institutional messages may appear more credible to sceptical readers who use language

quality as a heuristic. This strengthens the case for lure-specific training rather than generic "watch for bad grammar" advice that no longer matches experienced reality.

Deepfake voice and video add further channel complexity, though they are secondary to this thesis focus. The core argument remains: production quality lowers barriers to credibility, but emotional and social framing determines uptake.

Cognitive style and Norwegian empirical precedents

Tjostheim and Waterworth (2020) studied phishing susceptibility in a Norwegian representative sample, finding relationships between cognitive reflection, data-sharing willingness, and compliance with deceptive requests. Their 2022 follow-up work further supports cognitive style as a predictor. These studies establish an important national baseline: Norwegians are not uniformly naive. Many resist generic phishing when given time to reflect.

The present thesis extends this line by asking whether emotionally and socially framed lures bypass cognitive reflection differently. A student who scores high on analytical thinking may still respond to a message offering social belonging during a lonely semester, or to an urgent Altinn impersonation during tax season. Cognitive style and emotional framing may interact rather than compete as single predictors.

Including cognitive reflection items as controls in the planned survey allows comparison with prior Norwegian work while testing whether loneliness and lure type add explanatory power beyond cognitive style alone.

Student welfare, loneliness policy, and cybersecurity silos

Norwegian student welfare policy treats loneliness as a structural issue: campus design, student organisations, and health services (SHoT Study, 2022; Directorate of Health, 2024). Cybersecurity policy treats phishing as a technical and compliance issue: filters, awareness modules, and incident reporting (NSM, 2026). These domains rarely intersect in curriculum or institutional communication.

OsloMet initiatives such as structured social meeting programmes (OsloMet, 2024) address belonging explicitly. Security awareness modules address clicking behaviour. The thesis argues that institutional separation may miss a joint risk: socially meaningful lures

exploit the same unmet needs that welfare policy tries to reduce. Integration is not moralising but analytical—understanding where policies talk past each other.

Practical recommendations may therefore span departments: student welfare communications that mention fraud targeting lonely students; security teams that avoid shame-based messaging during high-stress periods such as exam weeks.

Planned methodology overview (Phase 2 and 3)

Although full methodology belongs to later thesis phases, ACIT5910 requires a concrete research plan. The planned design is exploratory sequential mixed methods (Braun & Clarke, 2006). Phase 2 administers an online survey with loneliness scales, scepticism items, situational pressure measures, demographic controls, and six vignettes (S1–S6): stranger prize scam; peer or campus event message; BankID/Altinn/university IT impersonation; urgent manager or exam office request; internship or job offer with time pressure; community or support invitation.

Respondents indicate intended behaviour on a fixed scale: click, verify via separate channel, ignore, or report. H1 compares social and belonging lures with stranger baselines across loneliness levels. H2 tests interaction between general scepticism and lure type. H3 models situational pressure as mediator. H4 tests verification habits and perceived non-punitive culture as moderators of coping appraisal.

Phase 3 interviews 12–18 participants sampled for variation in loneliness scores and vignette responses. Interview topics include: what made a message feel credible; emotional state at the time; knowledge-behaviour gaps; experiences of shame and reporting. Integration uses joint displays connecting statistical patterns with lived experience quotes.

Ethics: SIKT notification or approval before collection; informed consent; pseudonymisation; no live deceptive phishing without explicit approval; special care for fraud victimisation and shame topics. Data stored on institution-approved encrypted platforms with access limited to researcher and supervisor.

Limitations include self-report bias, convenience sampling, and cross-sectional design. Subgroup analysis by age may be required if

loneliness effects differ across cohorts as suggested by Wen et al. (2024). These limitations will be stated explicitly in Phase 2 and Phase 3 reporting.

Expected implications and academic positioning

If H1 and H2 are supported, awareness materials should stop treating scepticism as a single trait and address institutional and social impersonation explicitly. If H3 is supported, exam periods, tax seasons, and financial stress become predictable high-risk windows requiring targeted communication. If H4 is supported, culture and verification habits may matter as much as individual traits—organisations share responsibility for safer defaults and reporting support.

Academically, the thesis positions itself between human-factors cybersecurity, loneliness research, and Norwegian empirical security studies. It contributes a named paradox—the scepticism paradox—as an analytical tool rather than a metaphor. Peer review and process memo requirements in ACIT5910 will further refine argument structure, coherence, and audience awareness per writing workshop learning outcomes.

If hypotheses are not supported, the thesis will still advance discussion by documenting null findings in a Norwegian higher-education sample—valuable given the dominance of weakest-link assumptions in practice. Negative results would suggest revising the paradox or refining operationalisations rather than abandoning human-centred analysis.

Closing synthesis for Phase 1

This Phase 1 submission establishes that phishing in Norway cannot be explained solely by user ignorance or uniformly low scepticism. The literature supports a lure-specific, context-sensitive model in which emotional need, institutional trust, situational pressure, and security culture interact. The scepticism paradox names the central tension: rising caution toward strangers coexists with persistent vulnerability to socially and institutionally credible lures.

The planned mixed-methods work in Phase 2 and Phase 3 will test whether loneliness and lure-specific scepticism predict intended unsafe behaviour beyond cognitive style, and whether coping appraisal

factors including non-punitive culture reduce susceptibility. Until those data are collected, this document functions as the theoretical and empirical foundation required by ACIT5910: a critical literature review, clear problem statement, research questions, and a feasible research plan aligned with the master thesis title Beyond the Weakest Link: The Skepticism Paradox.

Peer review in ACIT5910 will stress-test clarity and argument flow before supervisor submission. The process memo will document how feedback on structure, APA formatting, and research question precision was incorporated. This iterative writing process mirrors the thesis itself: initial hypotheses and models are proposed, tested against evidence, and revised transparently rather than defended regardless of findings.

In sum, the weakest-link narrative asks the wrong question—who failed?—when the more productive question is under what lure-specific conditions do cautious people still respond unsafely, and which organisational practices make verification and reporting feasible? That reframing is the intellectual core of this master thesis and the practical reason Phase 1 matters beyond coursework requirements.

This document meets ACIT5910 Phase 1 scope by presenting a standalone Introduction and Literature Review that can evolve into the opening chapters of the full thesis. Terminology, hypotheses, and vignette design are aligned with the master project title and the problem statement approved in early supervision discussions, ensuring continuity across Phase 1, Phase 2, and Phase 3 rather than treating coursework and thesis as separate writing exercises.

References

- Alves, L. M., & Wilson, S. R. (2008). The effects of loneliness on telemarketing fraud vulnerability among older adults. *Journal of Elder Abuse & Neglect*, 20(1), 63–85.
- Bayl-Smith, P., Taib, R., Yu, K., & Wiggins, M. W. (2024). Response to a phishing attack: Persuasion and protection motivation in an organizational context. *Information and Computer Security*.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>

- Cacioppo, J. T., & Patrick, W. (2008). *Loneliness: Human nature and the need for social connection*. W. W. Norton.
- Chen, Y., et al. (2024). The effects of group discussion and role-playing training on self-efficacy, support-seeking, and reporting phishing emails. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*.
- Conteh, N. Y., & Schmick, P. J. (2016). Cybersecurity: Risk management framework and principles. In *Journal of Cyber Security Technology*.
- Cybersecurity and Infrastructure Security Agency. (n.d.). *Phishing guidance*. <https://www.cisa.gov/news-events/news/avoiding-social-engineering-and-phishing-attacks>
- Directorate of Health. (2024). Loneliness and public health.
- European Union Agency for Cybersecurity. (2024). *ENISA threat landscape 2024*. <https://www.enisa.europa.eu/publications/enisa-threat-landscape-2024>
- Finans Norge. (2026). *Self-defence against fraud (Selvforsvar mot svindel)*. <https://www.svindel.no/>
- Floyd, D. L., Prentice-Dunn, S., & Rogers, R. W. (2000). A meta-analysis of research on protection motivation theory. *Journal of Applied Social Psychology*, 30(2), 407–429.
- Frauenstein, E. D., & Flowerday, S. V. (2020). Susceptibility to phishing on social network sites. *Computers & Security*, 94, 101862.
- Hadnagy, C. (2018). *Social engineering: The science of human hacking* (2nd ed.). Wiley.
- Halcyon. (2026). *AI and ransomware threat report*.
- Jensen, T. H., et al. (2024). Remote work and cognitive bias in security decisions. *Journal of Information Security and Applications*.
- Khrono & Norwegian Student Organisation. (2024). Student loneliness reporting.
- Klüttsch, J., et al. (2024). Friend or phisher: How known senders and FOMO affect young adults' phishing susceptibility. *Humanities and Social Sciences Communications*, 11(1), 1–14.
- Microsoft. (2024). *Digital defense report*.

- Nettvet.no & Norwegian National Security Authority. (n.d.). Phishing guidance.
- Nkom. (n.d.). Digital fraud reports.
- Norwegian National Security Authority. (2026). *National cyber security risk assessment*.
- OsloMet. (2024). Student social connection initiatives.
- Parker, H. J., & Flowerday, S. V. (2020). Contributing factors to increased susceptibility to social media phishing attacks. *South African Journal of Information Management*, 22(1), a1176.
- Politiet. (2024). Online fraud statistics.
- Popovac, M., et al. (2020). Online risk perception and deception.
- Przybylski, A. K., et al. (2013). Motivational, emotional, and behavioral correlates of fear of missing out. *Computers in Human Behavior*, 29(4), 1841–1848.
- Rogers, R. W. (1975). A protection motivation theory of fear appeals and attitude change. *Journal of Psychology*, 91(1), 93–114.
- SHoT Study. (2022). Student health and wellbeing survey.
- Statistics Norway. (2025). Quality of life survey: Loneliness indicators.
- Tjostheim, I., & Waterworth, J. A. (2020). Phishing susceptibility in a Norwegian sample. *Proceedings of Norsk IKT-konferanse*.
- Tjostheim, I., & Waterworth, J. A. (2022). Cognitive reflection and phishing. *Proceedings of Norsk IKT-konferanse*.
- Trend Micro. (2026). *Phishing threat report*.
- Tsai, T. H., et al. (2021). Phishing and persuasion strategies. *Computers & Security*.
- Wen, Y., et al. (2022). Mechanisms linking loneliness and fraud.
- Wen, Y., et al. (2024). Loneliness and fraud susceptibility. *Journal of Consumer Affairs*.
- World Health Organization. (2023). Social connection and health.